CHAPTER 8

Hypothesis testing

8.1 Introduction

As in the previous chapter, let us suppose that a random sample from a population distribution, specified except for a vector of unknown parameters, is to be observed. However, rather than wishing to explicitly estimate the unknown parameters, let us now suppose that we are primarily concerned with using the resulting sample to test some particular hypothesis concerning them. As an illustration, suppose that a construction firm has just purchased a large supply of cables that have been guaranteed to have an average breaking strength of at least 7000 pounds per square inch (PSI). To verify this claim, the firm has decided to take a random sample of 10 of these cables to determine their breaking strengths. They will then use the result of this experiment to ascertain whether or not they accept the cable manufacturer's hypothesis that the population mean is at least 7000 PSI.

A statistical hypothesis is usually a statement about a set of parameters of a population distribution. It is called a hypothesis because it is not known whether or not it is true. A primary problem is to develop a procedure for determining whether or not the values of a random sample from this population are consistent with the hypothesis. For instance, consider a particular normally distributed population having an unknown mean value θ and known variance 1. The statement "θ is less than 1" is a statistical hypothesis that we could try to test by observing a random sample from this population. If the random sample is deemed to be consistent with the hypothesis under consideration, we say that the hypothesis has been "accepted"; otherwise we say that it has been "rejected."

Note that in accepting a given hypothesis we are not actually claiming that it is true but rather we are saying that the resulting data appear to be consistent with it. For instance, in the case of a normal $(\theta, 1)$ population, if a resulting sample of size 10 has an average value of 1.25, then although such a result cannot be regarded as being evidence in favor of the hypothesis "$\theta < 1$," it is not inconsistent with this hypothesis, which would thus be accepted. On the other hand, if the sample of size 10 has an average value of 3, then even though a sample value that large is possible when $\theta < 1$, it is so unlikely that it seems inconsistent with this hypothesis, which would thus be rejected.

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8.2 Significance levels

Consider a population having distribution F_θ, where θ is unknown, and suppose we want to test a specific hypothesis about θ. We shall denote this hypothesis by H_0 and call it the *null hypothesis*. For example, if F_θ is a normal distribution function with mean θ and variance equal to 1, then two possible null hypotheses about θ are

(a) $H_0 : \theta = 1$
(b) $H_0 : \theta \leq 1$

Thus the first of these hypotheses states that the population is normal with mean 1 and variance 1, whereas the second states that it is normal with variance 1 and a mean less than or equal to 1. Note that the null hypothesis in (a), when true, completely specifies the population distribution, whereas the null hypothesis in (b) does not. A hypothesis that, when true, completely specifies the population distribution is called a *simple* hypothesis; one that does not is called a *composite* hypothesis.

Suppose now that in order to test a specific null hypothesis H_0, a population sample of size n — say $X_1, \ldots, X_n$ — is to be observed. Based on these n values, we must decide whether or not to accept H_0. A test for H_0 can be specified by defining a region C in n-dimensional space with the proviso that the hypothesis is to be rejected if the random sample $X_1, \ldots, X_n$ turns out to lie in C and accepted otherwise. The region C is called the *critical region*. In other words, the statistical test determined by the critical region C is the one that

accepts H_0 if $(X_1, X_2, \ldots, X_n) \notin C$

and

rejects H_0 if $(X_1, \ldots, X_n) \in C$

For instance, a common test of the hypothesis that θ, the mean of a normal population with variance 1, is equal to 1 has a critical region given by

$$C = \{(X_1, \ldots, X_n) : |\bar{X} - 1| > 1.96/\sqrt{n}\} \tag{8.2.1}$$

Thus, this test calls for rejection of the null hypothesis that $\theta = 1$ when the sample average differs from 1 by more than 1.96 divided by the square root of the sample size.

It is important to note when developing a procedure for testing a given null hypothesis H_0 that, in any test, two different types of errors can result. The first of these, called a *type I error*, is said to result if the test incorrectly calls for rejecting H_0 when it is indeed correct. The second, called a *type II error*, results if the test calls for accepting H_0 when it is false.

Now, as was previously mentioned, the objective of a statistical test of H_0 is not to explicitly determine whether or not H_0 is true but rather to determine if its validity is consistent with the resultant data. Hence, with this objective it seems reasonable that H_0 should only be rejected if the resultant data are very unlikely when H_0 is true. The classical way of accomplishing this is to specify a value α and then require the test to have the property that whenever H_0 is true its probability of being rejected is never greater than α. The value α, called the *level of significance of the test*, is usually set in advance, with commonly chosen values being $\alpha = .1, .05, .005$. In other words, the classical approach to testing H_0 is to fix a significance level α and then require that the test have the property that the probability of a type I error occurring can never be greater than α.

Suppose now that we are interested in testing a certain hypothesis concerning θ, an unknown parameter of the population. Specifically, for a given set of parameter values w, suppose we are interested in testing

$$H_0 : \theta \in w$$

A common approach to developing a test of H_0, say at level of significance α, is to start by determining a point estimator of θ — say $d(\mathbf{X})$. The hypothesis is then rejected if $d(\mathbf{X})$ is "far away" from the region w. However, to determine how "far away" it need be to justify rejection of H_0, we need to determine the probability distribution of $d(\mathbf{X})$ when H_0 is true since this will usually enable us to determine the appropriate critical region so as to make the test have the required significance level α. For example, the test of the hypothesis that the mean of a normal $(\theta, 1)$ population is equal to 1, given by Equation (8.2.1), calls for rejection when the point estimate of θ — that is, the sample average — is farther than $1.96/\sqrt{n}$ away from 1. As we will see in the next section, the value $1.96/\sqrt{n}$ was chosen to meet a level of significance of $\alpha = .05$.

8.3 Tests concerning the mean of a normal population

8.3.1 Case of known variance

Suppose that $X_1, \ldots, X_n$ is a sample of size n from a normal distribution having an unknown mean μ and a known variance σ^2 and suppose we are interested in testing the null hypothesis

$$H_0 : \mu = \mu_0$$

against the alternative hypothesis

$$H_1 : \mu \neq \mu_0$$

where μ_0 is some specified constant.

Since $\overline{X} = \sum_{i=1}^{n} X_i/n$ is a natural point estimator of μ, it seems reasonable to accept H_0 if $\overline{X}$ is not too far from μ_0. That is, the critical region of the test would be of the form

$$C = \{X_1, \ldots, X_n : |\overline{X} - \mu_0| > c\} \tag{8.3.1}$$

for some suitably chosen value c.

If we desire that the test has significance level α, then we must determine the critical value c in Equation (8.3.1) that will make the type I error equal to α. That is, c must be such that

$$P_{\mu_0}\{|\overline{X} - \mu_0| > c\} = \alpha \tag{8.3.2}$$

where we write P_{μ_0} to mean that the preceding probability is to be computed under the assumption that $\mu = \mu_0$. However, when $\mu = \mu_0$, $\overline{X}$ will be normally distributed with mean μ_0 and variance σ^2/n and so Z, defined by

$$Z \equiv \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma}$$

will have a standard normal distribution. Now Equation (8.3.2) is equivalent to

$$P\left\{|Z| > \frac{c\sqrt{n}}{\sigma}\right\} = \alpha$$

or, equivalently,

$$2P\left\{Z > \frac{c\sqrt{n}}{\sigma}\right\} = \alpha$$

where Z is a standard normal random variable. However, we know that

$$P\{Z > z_{\alpha/2}\} = \alpha/2$$

and so

$$\frac{c\sqrt{n}}{\sigma} = z_{\alpha/2}$$

or

$$c = \frac{z_{\alpha/2}\sigma}{\sqrt{n}}$$

Thus, the significance level α test is to reject H_0 if $|\overline{X} - \mu_0| > z_{\alpha/2}\sigma/\sqrt{n}$ and accept otherwise; or, equivalently, to

$$\begin{aligned} &\text{reject} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}|\overline{X} - \mu_0| > z_{\alpha/2} \\ &\text{accept} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}|\overline{X} - \mu_0| \leq z_{\alpha/2} \end{aligned} \tag{8.3.3}$$

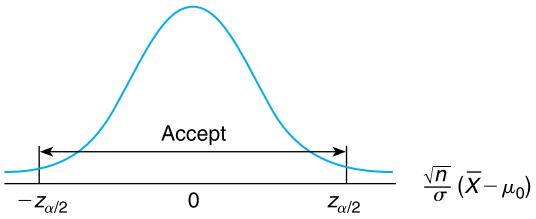


FIGURE 8.1

This can be pictorially represented as shown in Figure 8.1, where we have superimposed the standard normal density function [which is the density of the test statistic $\sqrt{n}(\overline{X} - \mu_0)/\sigma$ when H_0 is true].

Example 8.3.a. It is known that if a signal of value μ is sent from location A, then the value received at location B is normally distributed with mean μ and standard deviation 2. That is, the random noise added to the signal is an $N(0, 4)$ random variable. There is reason for the people at location B to suspect that the signal value $\mu = 8$ will be sent today. Test this hypothesis if the same signal value is independently sent five times and the average value received at location B is $\overline{X} = 9.5$.

Solution. Suppose we are testing at the 5 percent level of significance. To begin, we compute the test statistic

$$\frac{\sqrt{n}}{\sigma}|\overline{X} - \mu_0| = \frac{\sqrt{5}}{2}(1.5) = 1.68$$

Since this value is less than $z_{.025} = 1.96$, the hypothesis is accepted. In other words, the data are not inconsistent with the null hypothesis in the sense that a sample average as far from the value 8 as observed would be expected, when the true mean is 8, over 5 percent of the time. Note, however, that if a less stringent significance level were chosen — say $\alpha = .1$ — then the null hypothesis would have been rejected. This follows since $z_{.05} = 1.645$, which is less than 1.68. Hence, if we would have chosen a test that had a 10 percent chance of rejecting H_0 when H_0 was true, then the null hypothesis would have been rejected.

The "correct" level of significance to use in a given situation depends on the individual circumstances involved in that situation. For instance, if rejecting a null hypothesis H_0 would result in large costs that would thus be lost if H_0 were indeed true, then we might elect to be quite conservative and so choose a significance level of .05 or .01. Also, if we initially feel strongly that H_0 was correct, then we would require very stringent data evidence to the contrary for us to reject H_0. (That is, we would set a very low significance level in this situation.) ■

The test given by Equation (8.3.3) can be described as follows: For any observed value of the test statistic $\sqrt{n}|\overline{X} - \mu_0|/\sigma$, call it v, the test calls for rejection of

the null hypothesis if the probability that the test statistic would be as large as v when H_0 is true is less than or equal to the significance level α. From this, it follows that we can determine whether or not to accept the null hypothesis by computing, first, the value of the test statistic and, second, the probability that a standard normal would (in absolute value) exceed that quantity. This probability — called the p-value of the test — gives the critical significance level in the sense that H_0 will be accepted if the significance level α is less than the p-value and rejected if it is greater than or equal.

In practice, the significance level is often not set in advance but rather the data are looked at to determine the resultant p-value. Sometimes, this critical significance level is clearly much larger than any we would want to use, and so the null hypothesis can be readily accepted. At other times the p-value is so small that it is clear that the hypothesis should be rejected.

Example 8.3.b. In Example 8.3.a, suppose that the average of the 5 values received is $\overline{X} = 8.5$. In this case,

$$\frac{\sqrt{n}}{\sigma}|\overline{X} - \mu_0| = \frac{\sqrt{5}}{4} = .559$$

Since

$$\begin{aligned} P\{|Z| > .559\} &= 2P\{Z > .559\} \\ &= 2 \times .288 = .576 \end{aligned}$$

it follows that the p-value is .576 and thus the null hypothesis H_0 that the signal sent has value 8 would be accepted at any significance level $\alpha < .576$. Since we would clearly never want to test a null hypothesis using a significance level as large as .576, H_0 would be accepted.

On the other hand, if the average of the data values were 11.5, then the p-value of the test that the mean is equal to 8 would be

$$\begin{aligned} P\{|Z| > 1.75\sqrt{5}\} &= P\{|Z| > 3.913\} \\ &\approx .00005 \end{aligned}$$

For such a small p-value, the hypothesis that the value 8 was sent is rejected. ■

We have not yet talked about the probability of a type II error — that is, the probability of accepting the null hypothesis when the true mean μ is unequal to μ_0. This probability will depend on the value of μ, and so let us define $\beta(\mu)$ by

$$\beta(\mu) = P_\mu\{\text{acceptance of } H_0\}$$

$$= P_\mu \left\{ \left| \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} \right| \le z_{\alpha/2} \right\}$$
$$= P_\mu \left\{ -z_{\alpha/2} \le \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} \le z_{\alpha/2} \right\}$$

The function $\beta(\mu)$ is called the *operating characteristic* (or OC) *curve* and represents the probability that H_0 will be accepted when the true mean is μ.

To compute this probability, we use the fact that $\overline{X}$ is normal with mean μ and variance σ^2/n and so

$$Z \equiv \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim (0, 1)$$

Hence,

$$\begin{aligned}
\beta(\mu) &= P_\mu \left\{ -z_{\alpha/2} \le \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} \le z_{\alpha/2} \right\} \\
&= P_\mu \left\{ -z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \le \frac{\overline{X} - \mu_0 - \mu}{\sigma/\sqrt{n}} \le z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \right\} \\
&= P_\mu \left\{ -z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \le Z - \frac{\mu_0}{\sigma/\sqrt{n}} \le z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \right\} \\
&= P \left\{ \frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2} \le Z \le \frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2} \right\} \\
&= \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2} \right) - \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2} \right) \qquad (8.3.4)
\end{aligned}$$

where Φ is the standard normal distribution function.

For a fixed significance level α, the OC curve given by Equation (8.3.4) is symmetric about μ_0 and indeed will depend on μ only through $\sqrt{n}\,|\mu - \mu_0|/\sigma$. This curve with the abscissa changed from μ to $d = \sqrt{n}\,|\mu - \mu_0|/\sigma$ is presented in Figure 8.2 when $\alpha = .05$.

Example 8.3.c. For the problem presented in Example 8.3.a, let us determine the probability of accepting the null hypothesis that $\mu = 8$ when the actual value sent is 10. To do so, we compute

$$\frac{\sqrt{n}}{\sigma}(\mu_0 - \mu) = -\frac{\sqrt{5}}{2} \times 2 = -\sqrt{5}$$

As $z_{.025} = 1.96$, the desired probability is, from Equation (8.3.4),

$$\Phi(-\sqrt{5} + 1.96) - \Phi(-\sqrt{5} - 1.96)$$

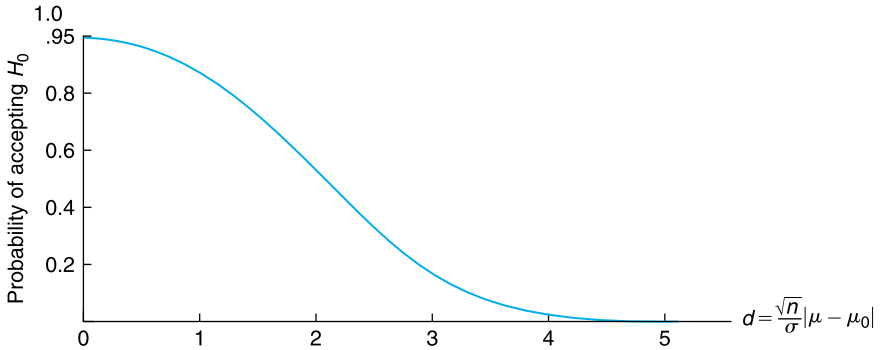


FIGURE 8.2
The OC curve for the two-sided normal test for significance level $\alpha = .05$.

$$= 1 - \Phi(\sqrt{5} - 1.96) - [1 - \Phi(\sqrt{5} + 1.96)]$$
$$= \Phi(4.196) - \Phi(.276)$$
$$= .392 \quad \blacksquare$$

Remark

The function $1 - \beta(\mu)$ is called the *power-function* of the test. Thus, for a given value μ, the power of the test is equal to the probability of rejection when μ is the true value. $\blacksquare$

The operating characteristic function is useful in determining how large the random sample need be to meet certain specifications concerning type II errors. For instance, suppose that we desire to determine the sample size n necessary to ensure that the probability of accepting $H_0 : \mu = \mu_0$ when the true mean is actually μ_1 is approximately β. That is, we want n to be such that

$$\beta(\mu_1) \approx \beta$$

But from Equation (8.3.4), this is equivalent to

$$\Phi\left(\frac{\sqrt{n}(\mu_0 - \mu_1)}{\sigma} + z_{\alpha/2}\right) - \Phi\left(\frac{\sqrt{n}(\mu_0 - \mu_1)}{\sigma} - z_{\alpha/2}\right) \approx \beta \tag{8.3.5}$$

Although the foregoing cannot be analytically solved for n, a solution can be obtained by using the standard normal distribution table. In addition, an approximation for n can be derived from Equation (8.3.5) as follows. To start, suppose that $\mu_1 > \mu_0$. Then, because this implies that

$$\frac{\sqrt{n}(\mu_0 - \mu_1)}{\sigma} - z_{\alpha/2} \leq -z_{\alpha/2}$$

it follows, since Φ is an increasing function, that

$$\Phi\left(\frac{\sqrt{n}(\mu_0-\mu_1)}{\sigma}-z_{\alpha/2}\right)\le\Phi(-z_{\alpha/2})=P\{Z\le -z_{\alpha/2}\}=P\{Z\ge z_{\alpha/2}\}=\alpha/2$$

Hence, we can take

$$\Phi\left(\frac{\sqrt{n}(\mu_0-\mu_1)}{\sigma}-z_{\alpha/2}\right)\approx 0$$

and so from Equation (8.3.5)

$$\beta\approx\Phi\left(\frac{\sqrt{n}(\mu_0-\mu_1)}{\sigma}+z_{\alpha/2}\right) \tag{8.3.6}$$

or, since

$$\beta=P\{Z>z_\beta\}=P\{Z<-z_\beta\}=\Phi(-z_\beta)$$

we obtain from Equation (8.3.6) that

$$-z_\beta\approx(\mu_0-\mu_1)\frac{\sqrt{n}}{\sigma}+z_{\alpha/2}$$

or

$$n\approx\frac{(z_{\alpha/2}+z_\beta)^2\sigma^2}{(\mu_1-\mu_0)^2} \tag{8.3.7}$$

In fact, the same approximation would result when $\mu_1<\mu_0$ (the details are left as an exercise) and so Equation (8.3.7) is in all cases a reasonable approximation to the sample size necessary to ensure that the type II error at the value $\mu=\mu_1$ is approximately equal to β.

Example 8.3.d. For the problem of Example 8.3.a, how many signals need be sent so that the .05 level test of $H_0:\mu=8$ has at least a 75 percent probability of rejection when $\mu=9.2$?

Solution. Since $z_{.025}=1.96$, $z_{.25}=.67$, the approximation from Equation (8.3.7) yields

$$n\approx\frac{(1.96+.67)^2}{(1.2)^2}4=19.21$$

Hence a sample of size 20 is needed. From Equation (8.3.4), we see that with $n=20$

$$\begin{aligned}\beta(9.2)&=\Phi\left(-\frac{1.2\sqrt{20}}{2}+1.96\right)-\Phi\left(-\frac{1.2\sqrt{20}}{2}-1.96\right)\\&=\Phi(-.723)-\Phi(-4.643)\end{aligned}$$

$$\approx 1 - \Phi(.723)$$
$$\approx .235$$

Therefore, if the message is sent 20 times, then there is a 76.5 percent chance that the null hypothesis $\mu = 8$ will be rejected when the true mean is 9.2. ■

8.3.1.1 One-sided tests

In testing the null hypothesis that $\mu = \mu_0$, we have chosen a test that calls for rejection when $\overline{X}$ is far from μ_0. That is, a very small value of $\overline{X}$ or a very large value appears to make it unlikely that μ (which $\overline{X}$ is estimating) could equal μ_0. However, what happens when the only alternative to μ being equal to μ_0 is for μ to be greater than μ_0? That is, what happens when the alternative hypothesis to $H_0 : \mu = \mu_0$ is $H_1 : \mu > \mu_0$? Clearly, in this latter case we would not want to reject H_0 when $\overline{X}$ is small (since a small $\overline{X}$ is more likely when H_0 is true than when H_1 is true). Thus, in testing

$$H_0 : \mu = \mu_0 \qquad \text{versus} \qquad H_1 : \mu > \mu_0 \tag{8.3.8}$$

we should reject H_0 when $\overline{X}$, the point estimate of μ_0, is much greater than μ_0. That is, the critical region should be of the following form:

$$C = \{(X_1, \ldots, X_n) : \overline{X} - \mu_0 > c\}$$

Since the probability of rejection should equal α when H_0 is true (that is, when $\mu = \mu_0$), we require that c be such that

$$P_{\mu_0}\{\overline{X} - \mu_0 > c\} = \alpha \tag{8.3.9}$$

But since

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma}$$

has a standard normal distribution when H_0 is true, Equation (8.3.9) is equivalent to

$$P\left\{Z > \frac{c\sqrt{n}}{\sigma}\right\} = \alpha$$

when Z is a standard normal. But since

$$P\{Z > z_\alpha\} = \alpha$$

we see that

$$c = \frac{z_\alpha \sigma}{\sqrt{n}}$$

Hence, the test of the Hypothesis 8.3.8 is to reject H_0 if $\overline{X} - \mu_0 > z_\alpha \sigma/\sqrt{n}$, and accept otherwise; or, equivalently, to

$$\begin{aligned} &\text{accept} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}(\overline{X} - \mu_0) \le z_\alpha \\ &\text{reject} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}(\overline{X} - \mu_0) > z_\alpha \end{aligned} \tag{8.3.10}$$

This is called a *one-sided* critical region (since it calls for rejection only when $\overline{X}$ is large). Correspondingly, the hypothesis testing problem

$$H_0 : \mu = \mu_0$$
$$H_1 : \mu > \mu_0$$

is called a one-sided testing problem (in contrast to the *two-sided* problem that results when the alternative hypothesis is $H_1 : \mu \neq \mu_0$).

To compute the p-value in the one-sided test, Equation (8.3.10), we first use the data to determine the value of the statistic $\sqrt{n}(\overline{X} - \mu_0)/\sigma$. The p-value is then equal to the probability that a standard normal would be at least as large as this value.

Example 8.3.e. Suppose in Example 8.3.a that we know in advance that the signal value is at least as large as 8. What can be concluded in this case?

Solution. To see if the data are consistent with the hypothesis that the mean is 8, we test

$$H_0 : \mu = 8$$

against the one-sided alternative

$$H_1 : \mu > 8$$

The value of the test statistic is $\sqrt{n}(\overline{X} - \mu_0)/\sigma = \sqrt{5}(9.5 - 8)/2 = 1.68$, and the p-value is the probability that a standard normal would exceed 1.68, namely,

$$p\text{-value} = 1 - \Phi(1.68) = .0465$$

The test would call for rejection at all significance levels greater than or equal to .0465. It would, for instance, reject the null hypothesis at the $\alpha = .05$ level of significance. ■

The operating characteristic function of the one-sided test, Equation (8.3.10),

$$\beta(\mu) = P_\mu\{\text{accepting } H_0\}$$

can be obtained as follows:

$$\begin{aligned}\beta(\mu) &= P_\mu\left\{\overline{X} \le \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}}\right\} \\ &= P\left\{\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \le \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma} + z_\alpha\right\} \\ &= P\left\{Z \le \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma} + z_\alpha\right\}, \quad Z \sim (0, 1)\end{aligned}$$

where the last equation follows since $\sqrt{n}(\overline{X} - \mu)/\sigma$ has a standard normal distribution. Hence we can write

$$\beta(\mu) = \Phi\left(\frac{\sqrt{n}(\mu_0 - \mu)}{\sigma} + z_\alpha\right)$$

Since Φ, being a distribution function, is increasing in its argument, it follows that $\beta(\mu)$ decreases in μ, which is intuitively pleasing since it certainly seems reasonable that the larger the true mean μ, the less likely it should be to conclude that $\mu \le \mu_0$. Also since $\Phi(z_\alpha) = 1 - \alpha$, it follows that

$$\beta(\mu_0) = 1 - \alpha$$

The test given by Equation (8.3.10), which was designed to test $H_0 : \mu = \mu_0$ versus $H_1 : \mu > \mu_0$, can also be used to test, at level of significance α, the one-sided hypothesis

$$H_0 : \mu \le \mu_0$$

versus

$$H_1 : \mu > \mu_0$$

To verify that it remains a level α test, we need to show that the probability of rejection is never greater than α when H_0 is true. That is, we must verify that

$$1 - \beta(\mu) \le \alpha \qquad \text{for all } \mu \le \mu_0$$

or

$$\beta(\mu) \ge 1 - \alpha \qquad \text{for all } \mu \le \mu_0$$

But it has previously been shown that for the test given by Equation (8.3.10), $\beta(\mu)$ decreases in μ and $\beta(\mu_0) = 1 - \alpha$. This gives that

$$\beta(\mu) \ge \beta(\mu_0) = 1 - \alpha \qquad \text{for all } \mu \le \mu_0$$

which shows that the test given by Equation (8.3.10) remains a level α test for $H_0 : \mu \le \mu_0$ against the alternative hypothesis $H_1 : \mu > \mu_0$.

Remark

We can also test the one-sided hypothesis

$$H_0 : \mu = \mu_0 \quad (\text{or } \mu \geq \mu_0) \qquad \text{versus} \qquad H_1 : \mu < \mu_0$$

at significance level α by

$$\text{accepting} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}(\overline{X} - \mu_0) \geq -z_\alpha$$
$$\text{rejecting} \quad H_0 \quad \text{otherwise}$$

This test can alternatively be performed by first computing the value of the test statistic $\sqrt{n}(\overline{X} - \mu_0)/\sigma$. The p-value would then equal the probability that a standard normal would be less than this value, and the hypothesis would be rejected at any significance level greater than or equal to this p-value.

Example 8.3.f. All cigarettes presently on the market have an average nicotine content of at least 1.6 mg per cigarette. A firm that produces cigarettes claims that it has discovered a new way to cure tobacco leaves that will result in the average nicotine content of a cigarette being less than 1.6 mg. To test this claim, a sample of 20 of the firm's cigarettes were analyzed. If it is known that the standard deviation of a cigarette's nicotine content is .8 mg, what conclusions can be drawn, at the 5 percent level of significance, if the average nicotine content of the 20 cigarettes is 1.54?

Note: The above raises the question of how we would know in advance that the standard deviation is .8. One possibility is that the variation in a cigarette's nicotine content is due to variability in the amount of tobacco in each cigarette and not on the method of curing that is used. Hence, the standard deviation can be known from previous experience.

Solution. We must first decide on the appropriate null hypothesis. As was previously noted, our approach to testing is not symmetric with respect to the null and the alternative hypotheses since we consider only tests having the property that their probability of rejecting the null hypothesis when it is true will never exceed the significance level α. Thus, whereas rejection of the null hypothesis is a strong statement about the data not being consistent with this hypothesis, an analogous statement cannot be made when the null hypothesis is accepted. Hence, since in the preceding example we would like to endorse the producer's claims only when there is substantial evidence for it, we should take this claim as the alternative hypothesis. That is, we should test

$$H_0 : \mu \geq 1.6 \qquad \text{versus} \qquad H_1 : \mu < 1.6$$

Now, the value of the test statistic is

$$\sqrt{n}(\overline{X} - \mu_0)/\sigma = \sqrt{20}(1.54 - 1.6)/.8 = -.336$$

and so the p-value is given by

$$\begin{aligned} p\text{-value} &= P\{Z < -.336\}, \qquad Z \sim N(0,1) \\ &= .368 \end{aligned}$$

Since this value is greater than .05, the foregoing data do not enable us to reject, at the .05 percent level of significance, the hypothesis that the mean nicotine content exceeds 1.6 mg. In other words, the evidence, although supporting the cigarette producer's claim, is not strong enough to prove that claim. ■

Remarks

(a) There is a direct analogy between confidence interval estimation and hypothesis testing. For instance, for a normal population having mean μ and known variance σ^2, we have shown in Section 7.3 that a $100(1-\alpha)$ percent confidence interval for μ is given by

$$\mu \in \left(\overline{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \overline{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

where $\overline{x}$ is the observed sample mean. More formally, the preceding confidence interval statement is equivalent to

$$P\left\{\mu \in \left(\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)\right\} = 1-\alpha$$

Hence, if $\mu = \mu_0$, then the probability that μ_0 will fall in the interval

$$\left(\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

is $1-\alpha$, implying that a significance level α test of $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$ is to reject H_0 when

$$\mu_0 \notin \left(\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

Similarly, since a $100(1-\alpha)$ percent one-sided confidence interval for μ is given by

$$\mu \in \left(\overline{X} - z_{\alpha}\frac{\sigma}{\sqrt{n}}, \infty\right)$$

it follows that an α-level significance test of $H_0 : \mu \leq \mu_0$ versus $H_1 : \mu > \mu_0$ is to reject H_0 when $\mu_0 \notin (\overline{X} - z_\alpha\sigma/\sqrt{n}, \infty)$ — that is, when $\mu_0 < \overline{X} - z_\alpha\sigma/\sqrt{n}$.

(b) A remark on robustness A test that performs well even when the underlying assumptions on which it is based are violated is said to be *robust*. For

Table 8.1 $X_1, \ldots, X_n$ Is a Sample from a (μ, σ^2) Population, σ^2 Is Known, $\overline{X} = \sum_{i=1}^{n} X_i/n$.

H_0	H_1	Test Statistic TS	Significance Level α Test	p-Value if $TS = t$
$\mu = \mu_0$	$\mu \neq \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/\sigma$	Reject if $\|TS\| > z_{\alpha/2}$	$2P\{Z \geq \|t\|\}$
$\mu \leq \mu_0$	$\mu > \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/\sigma$	Reject if $TS > z_\alpha$	$P\{Z \geq t\}$
$\mu \geq \mu_0$	$\mu < \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/\sigma$	Reject if $TS < -z_\alpha$	$P\{Z \leq t\}$

Z is a standard normal random variable.

instance, the tests of Sections 8.3.1 and 8.3.1.1 were derived under the assumption that the underlying population distribution is normal with known variance σ^2. However, in deriving these tests, this assumption was used only to conclude that $\overline{X}$ also has a normal distribution. But, by the central limit theorem, it follows that for a reasonably large sample size, $\overline{X}$ will approximately have a normal distribution no matter what the underlying distribution. Thus we can conclude that these tests will be relatively robust for any population distribution with variance σ^2.

Table 8.1 summarizes the tests of this subsection.

8.3.2 Case of unknown variance: the t-test

Up to now we have supposed that the only unknown parameter of the normal population distribution is its mean. However, the more common situation is one where the mean μ and variance σ^2 are both unknown. Let us suppose this to be the case and again consider a test of the hypothesis that the mean is equal to some specified value μ_0. That is, consider a test of

$$H_0 : \mu = \mu_0$$

versus the alternative

$$H_1 : \mu \neq \mu_0$$

It should be noted that the null hypothesis is not a simple hypothesis since it does not specify the value of σ^2.

As before, it seems reasonable to reject H_0 when the sample mean $\overline{X}$ is far from μ_0. However, how far away it need be to justify rejection will depend on the variance σ^2. Recall that when the value of σ^2 was known, the test called for rejecting H_0 when $|\overline{X} - \mu_0|$ exceeded $z_{\alpha/2}\sigma/\sqrt{n}$ or, equivalently, when

$$\left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \right| > z_{\alpha/2}$$

Now when σ^2 is no longer known, it seems reasonable to estimate it by

$$S^2 = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{n-1}$$

and then to reject H_0 when

$$\left| \frac{\overline{X} - \mu_0}{S/\sqrt{n}} \right|$$

is large.

To determine how large a value of the statistic

$$\left| \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} \right|$$

to require for rejection, in order that the resulting test have significance level α, we must determine the probability distribution of this statistic when H_0 is true. However, as shown in Section 6.5, the statistic T, defined by

$$T = \frac{\sqrt{n}(\overline{X} - \mu_0)}{S}$$

has, when $\mu = \mu_0$, a t-distribution with $n-1$ degrees of freedom. Hence,

$$P_{\mu_0}\left\{ -t_{\alpha/2,n-1} \le \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} \le t_{\alpha/2,n-1} \right\} = 1 - \alpha \tag{8.3.11}$$

where $t_{\alpha/2,n-1}$ is the 100 $\alpha/2$ upper percentile value of the t-distribution with $n-1$ degrees of freedom. (That is, $P\{T_{n-1} \ge t_{\alpha/2,n-1}\} = P\{T_{n-1} \le -t_{\alpha/2,n-1}\} = \alpha/2$ when T_{n-1} has a t-distribution with $n-1$ degrees of freedom.) From Equation (8.3.11) we see that the appropriate significance level α test of

$$H_0 : \mu = \mu_0 \qquad \text{versus} \qquad H_1 : \mu \ne \mu_0$$

is, when σ^2 is unknown, to

$$\begin{aligned} &\text{accept} \quad H_0 \quad \text{if} \quad \left| \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} \right| \le t_{\alpha/2,\, n-1} \\ &\text{reject} \quad H_0 \quad \text{if} \quad \left| \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} \right| > t_{\alpha/2,\, n-1} \end{aligned} \tag{8.3.12}$$

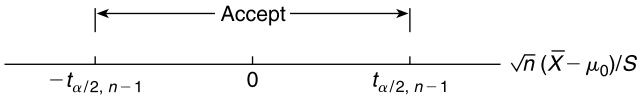


FIGURE 8.3
The two-sided t-test.

The test defined by Equation (8.3.12) is called a *two-sided t-test*. It is pictorially illustrated in Figure 8.3.

If we let t denote the observed value of the test statistic $T = \sqrt{n}(\overline{X} - \mu_0)/S$, then the p-value of the test is the probability that $|T|$ would exceed $|t|$ when H_0 is true. That is, the p-value is the probability that the absolute value of a t-random variable with $n - 1$ degrees of freedom would exceed $|t|$. The test then calls for rejection at all significance levels higher than the p-value and acceptance at all lower significance levels.

Example 8.3.g. Among a clinic's patients having blood cholesterol levels ranging in the medium to high range (at least 220 milliliters per deciliter of serum), volunteers were recruited to test a new drug designed to reduce blood cholesterol. A group of 50 volunteers was given the drug for 1 month and the changes in their blood cholesterol levels were noted. If the average change was a reduction of 14.8 with a sample standard deviation of 6.4, what conclusions can be drawn?

Solution. Let us start by testing the hypothesis that the change could be due solely to chance — that is, that the 50 changes constitute a normal sample with mean 0. Because the value of the t-statistic used to test the hypothesis that a normal mean is equal to 0 is

$$T = \sqrt{n}\,\overline{X}/S = \sqrt{50}\,14.8/6.4 = 16.352$$

it clear that we should reject the hypothesis that the changes were solely due to chance. Unfortunately, however, we are not justified at this point in concluding that the changes were due to the specific drug used and not to some other possibility. For instance, it is well known that any medication received by a patient (whether or not this medication is directly relevant to the patient's suffering) often leads to an improvement in the patient's condition — the so-called placebo effect. Also, another possibility that may need to be taken into account would be the weather conditions during the month of testing, for it is certainly conceivable that this affects blood cholesterol level. Indeed, it must be concluded that the foregoing was a very poorly designed experiment, for in order to test whether a specific treatment has an effect on a disease that may be affected by many things, we should try to design the experiment so as to neutralize all other possible causes. The accepted approach for accomplishing this is to divide the volunteers at random into two groups — one group to receive the drug and the other to receive a placebo (that is, a tablet that looks and tastes

like the actual drug but has no physiological effect). The volunteers should not be told whether they are in the actual or control group, and indeed it is best if even the clinicians do not have this information (the so-called double-blind test) so as not to allow their own biases to play a role. Since the two groups are chosen at random from among the volunteers, we can now hope that on average all factors affecting the two groups will be the same except that one received the actual drug and the other a placebo. Hence, any difference in performance between the groups can be attributed to the drug. ■

Example 8.3.h. A public health official claims that the mean home water use is 350 gallons a day. To verify this claim, a study of 20 randomly selected homes was instigated with the result that the average daily water uses of these 20 homes were as follows:

340	344	362	375
356	386	354	364
332	402	340	355
362	322	372	324
318	360	338	370

Do the data contradict the official's claim?

Solution. To determine if the data contradict the official's claim, we need to test

$$H_0 : \mu = 350 \qquad \text{versus} \qquad H_1 : \mu \neq 350$$

This can be accomplished by noting first that the sample mean and sample standard deviation of the preceding data set are

$$\overline{X} = 353.8, \qquad S = 21.8478$$

Thus, the value of the test statistic is

$$T = \frac{\sqrt{20}(3.8)}{21.8478} = .7778$$

Because this is less than $t_{.05,19} = 1.730$, the null hypothesis is accepted at the 10 percent level of significance. Indeed, the p-value of the test data is

$$p\text{-value} = P\{|T_{19}| > .7778\} = 2P\{T_{19} > .7778\} = .4462$$

indicating that the null hypothesis would be accepted at any reasonable significance level, and thus that the data are not inconsistent with the claim of the health official. ■

R can be used to perform the computations in a t-test. To test the null hypothesis that a mean μ is equal to μ_0 against the alternative that $\mu \neq \mu_0$ when the

data are $x_1, \ldots, x_n$, type

$>x = c(x_1, \ldots, x_n)$
$>$t.test(x, mu $= \mu_0$)

and hit return. Upon doing so, the output will include both a 95 percent confidence interval and the p-value. For instance, to use R to solve Example 8.3.h, do the following:

$>x = c(340, 356, 332, 362, 318, 344, 386, 402, 322, 360, 362, 354, 340, 372, 338,$
$375, 363, 355, 324, 370)$
$>$t.test(x, mu $= 350$)

After hitting return after the second line, you will receive the following output:

One Sample t-test
data: x
t = 0.77784, df = 19, p-value = 0.4462
alternative hypothesis: true mean is not equal to 350
95 percent confidence interval:
343.5749 364.0251
sample estimates:
mean of x
353.8

We can use a one-sided t-test to test the hypothesis

$H_0 : \mu = \mu_0 \qquad (\text{or } H_0 : \mu \leq \mu_0)$

against the one-sided alternative

$H_1 : \mu > \mu_0$

The significance level α test is to

$$\begin{aligned} &\text{accept} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} \leq t_{\alpha, n-1} \\ &\text{reject} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} > t_{\alpha, n-1} \end{aligned} \tag{8.3.13}$$

If $\sqrt{n}(\overline{X} - \mu_0)/S = v$, then the p-value of the test is the probability that a t-random variable with $n - 1$ degrees of freedom would be at least as large as v.

The significance level α test of

$H_0 : \mu = \mu_0 \qquad (\text{or } H_0 : \mu \geq \mu_0)$

versus the alternative

$$H_1 : \mu < \mu_0$$

is to

$$\text{accept} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} \geq -t_{\alpha, n-1}$$

$$\text{reject} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} < -t_{\alpha, n-1}$$

The p-value of this test is the probability that a t-random variable with $n - 1$ degrees of freedom would be less than or equal to the observed value of $\sqrt{n}(\overline{X} - \mu_0)/S$.

Example 8.3.i. The manufacturer of a new fiberglass tire claims that its average life will be at least 40,000 miles. To verify this claim a sample of 12 tires is tested, with their lifetimes (in 1000s of miles) being as follows:

Tire	1	2	3	4	5	6	7	8	9	10	11	12
Life	36.1	40.2	33.8	38.5	42	35.8	37	41	36.8	37.2	33	36

Test the manufacturer's claim at the 5 percent level of significance.

Solution. To determine whether the foregoing data are consistent with the hypothesis that the mean life is at least 40,000 miles, we will test

$$H_0 : \mu \geq 40{,}000 \qquad \text{versus} \qquad H_1 : \mu < 40{,}000$$

A computation gives that

$$\overline{X} = 37.2833, \qquad S = 2.7319$$

and so the value of the test statistic is

$$T = \frac{\sqrt{12}(37.2833 - 40)}{2.7319} = -3.4448$$

Since this is less than $-t_{.05,11} = -1.796$, the null hypothesis is rejected at the 5 percent level of significance. Indeed, the p-value of the test data is

$$p\text{-value} = P\{T_{11} < -3.4448\} = P\{T_{11} > 3.4448\} = .0027$$

indicating that the manufacturer's claim would be rejected at any significance level greater than .003. ■

The preceding could also have been solved by using R. R performs one-sided t-tests by specifying that the alternative hypothesis is that the mean is either less than or greater than μ_0. That is, to test

$H_0 : \mu = \mu_0$ versus $H_1 : \mu > \mu_0$

when using data $x_1, \ldots, x_n$, just type

$>x = c(x_1, \ldots, x_n)$
$>$t.test(x, alternative =“greater”, mu = μ_0)

and hit return. Thus, for Example 8.3.i, we obtain

$>x = c(36.1, 40.2, 33.8, 38.5, 42, 35.8, 37, 41, 36.8, 37.2, 33, 36)$
$>$t.test(x, alternative = “less”, mu = 40)

One Sample t-test
data: x
t = -3.4448, df = 11, p-value = 0.002739
alternative hypothesis: true mean is less than 40
95 percent confidence interval:
-Inf 38.69963
sample estimates:
mean of x
37.28333

Example 8.3.j. In a single-server queueing system in which customers arrive according to a Poisson process, the long-run average queueing delay per customer depends on the service distribution through its mean and variance. Indeed, if μ is the mean service time, and σ^2 is the variance of a service time, then the average amount of time that a customer spends waiting in queue is given by

$$\frac{\lambda(\mu^2 + \sigma^2)}{2(1 - \lambda\mu)}$$

provided that $\lambda\mu < 1$, where λ is the arrival rate. (The average delay is infinite if $\lambda\mu \geq 1$.) As can be seen by this formula, the average delay is quite large when μ is only slightly smaller than $1/\lambda$, where, since λ is the arrival *rate*, $1/\lambda$ is the average time between arrivals.

Suppose that the owner of a service station will hire a second server if it can be shown that the average service time exceeds 8 minutes. The following data give the service times (in minutes) of 28 customers of this queueing system. Do they indicate that the mean service time is greater than 8 minutes?

8.6, 9.4, 5.0, 4.4, 3.7, 11.4, 10.0, 7.6, 14.4, 12.2, 11.0, 14.4, 9.3, 10.5,
10.3, 7.7, 8.3, 6.4, 9.2, 5.7, 7.9, 9.4, 9.0, 13.3, 11.6, 10.0, 9.5, 6.6

16. Suppose that team A and team B are to play a National Football League game and that team A is favored by f points. Let $S(A)$ and $S(B)$ denote the scores of teams A and B, and let $X = S(A) - S(B) - f$. That is, X is the amount by which team A beats the point spread. It has been claimed that the distribution of X is normal with mean 0 and standard deviation 14. Use data from randomly chosen football games to test this hypothesis.
17. A medical scientist believes that the average basal temperature of (outwardly) healthy individuals has increased over time and is now greater than 98.6 degrees Fahrenheit (37 degrees Celsius). To prove this, she has randomly selected 100 healthy individuals. If their mean temperature is 98.74 with a sample standard deviation of 1.1 degrees, does this prove her claim at the 5 percent level? What about at the 1 percent level?
18. Use the results of a Sunday's worth of NFL professional football games to test the hypothesis that the average number of points scored by winning teams is less than or equal to 28. Use the 5 percent level of significance.
19. Use the results of a Sunday's worth of major league baseball scores to test the hypothesis that the average number of runs scored by winning teams is at least 5.6. Use the 5 percent level of significance.
20. A car is advertised as having a gas mileage rating of at least 30 miles/gallon in highway driving. If the miles per gallon obtained in 10 independent experiments are 26, 24, 20, 25, 27, 25, 28, 30, 26, 33, should you believe the advertisement? What assumptions are you making?
21. A producer specifies that the mean lifetime of a certain type of battery is at least 240 hours. A sample of 18 such batteries yielded the following data.

237	242	232
242	248	230
244	243	254
262	234	220
225	236	232
218	228	240

Assuming that the life of the batteries is approximately normally distributed, do the data indicate that the specifications are not being met?
22. Use the data of Example 2.3.i of Chapter 2 to test the null hypothesis that the average noise level directly outside of Grand Central Station is less than or equal to 80 decibels.
23. An oil company claims that the sulfur content of its diesel fuel is at most .15 percent. To check this claim, the sulfur contents of 40 randomly chosen samples were determined; the resulting sample mean and sample standard deviation were .162 and .040. Using the 5 percent level of significance, can we conclude that the company's claims are invalid?
24. A company supplies plastic sheets for industrial use. A new type of plastic has been produced and the company would like to claim that the average

stress resistance of this new product is at least 30.0, where stress resistance is measured in pounds per square inch (psi) necessary to crack the sheet. The following random sample was drawn off the production line. Based on this sample, would the claim clearly be unjustified?

30.1	32.7	22.5	27.5
27.7	29.8	28.9	31.4
31.2	24.3	26.4	22.8
29.1	33.4	32.5	21.7

Assume normality and use the 5 percent level of significance.

25. It is claimed that a certain type of bipolar transistor has a mean value of current gain that is at least 210. A sample of these transistors is tested. If the sample mean value of current gain is 200 with a sample standard deviation of 35, would the claim be rejected at the 5 percent level of significance if
    a. the sample size is 25;
    b. the sample size is 64?
26. A manufacturer of capacitors claims that the breakdown voltage of these capacitors has a mean value of at least 100 V. A test of 12 of these capacitors yielded the following breakdown voltages:

    96, 98, 105, 92, 111, 114, 99, 103, 95, 101, 106, 97

    Do these results prove the manufacturer's claim? Do they disprove them?
27. A sample of 10 fish were caught at lake A and their PCB concentrations were measured using a certain technique. The resulting data in parts per million were

    Lake A: 11.5, 10.8, 11.6, 9.4, 12.4, 11.4, 12.2, 11, 10.6, 10.8

    In addition, a sample of 8 fish were caught at lake B and their levels of PCB were measured by a different technique than that used at lake A. The resultant data were

    Lake B: 11.8, 12.6, 12.2, 12.5, 11.7, 12.1, 10.4, 12.6

    If it is known that the measuring technique used at lake A has a variance of .09 whereas the one used at lake B has a variance of .16, could you reject (at the 5 percent level of significance) a claim that the two lakes are equally contaminated?
28. A method for measuring the pH level of a solution yields a measurement value that is normally distributed with a mean equal to the actual pH of the solution and with a standard deviation equal to .05. An environmental pollution scientist claims that two different solutions come from the same source. If this were so, then the pH level of the solutions would

be equal. To test the plausibility of this claim, 10 independent measurements were made of the pH level for both solutions, with the following data resulting.

Measurements of Solution A	Measurements of Solution B
6.24	6.27
6.31	6.25
6.28	6.33
6.30	6.27
6.25	6.24
6.26	6.31
6.24	6.28
6.29	6.29
6.22	6.34
6.28	6.27

a. Do the data disprove the scientist's claim? Use the 5 percent level of significance.
b. What is the p-value?

29. The following are the values of independent samples from two different populations.

Sample 1	122, 114, 130, 165, 144, 133, 139, 142, 150
Sample 2	108, 125, 122, 140, 132, 120, 137, 128, 138

Let μ_1 and μ_2 be the respective means of the two populations. Find the p-value of the test of the null hypothesis

$$H_0 : \mu_1 \leq \mu_2$$

versus the alternative

$$H_1 : \mu_1 > \mu_2$$

when the population standard deviations are $\sigma_1 = 10$ and **(a)** $\sigma_2 = 5$; **(b)** $\sigma_2 = 10$; **(c)** $\sigma_2 = 20$.

30. The data below give the lifetimes in hundreds of hours of samples of two types of electronic tubes. Past lifetime data of such tubes have shown that they can often be modeled as arising from a lognormal distribution. That is, the logarithms of the data are normally distributed. Assuming that variance of the logarithms is equal for the two populations, test, at the 5 percent level of significance, the hypothesis that the two population distributions are identical.

Type 1	32, 84, 37, 42, 78, 62, 59, 74
Type 2	39, 111, 55, 106, 90, 87, 85

31. The viscosity of two different brands of car oil is measured and the following data resulted:

Brand 1	10.62, 10.58, 10.33, 10.72, 10.44, 10.74
Brand 2	10.50, 10.52, 10.58, 10.62, 10.55, 10.51, 10.53

Test the hypothesis that the mean viscosity of the two brands is equal, assuming that the populations have normal distributions with equal variances.

32. It is argued that the resistance of wire A is greater than the resistance of wire B. You make tests on each wire with the following results.

Wire A	Wire B
.140 ohm	.135 ohm
.138	.140
.143	.136
.142	.142
.144	.138
.137	.140

What conclusion can you draw at the 10 percent significance level? Explain what assumptions you are making.

In Problems 33 through 40, assume that the population distributions are normal and have equal variances.

33. Twenty-five men between the ages of 25 and 30, who were participating in a well-known heart study carried out in Framingham, Massachusetts, were randomly selected. Of these, 11 were smokers and 14 were not. The following data refer to readings of their systolic blood pressure.

Smokers	Nonsmokers
124	130
134	122
136	128
125	129
133	118
127	122
135	116
131	127
133	135
125	120
118	122
	120
	115
	123

Use these data to test the hypothesis that the mean blood pressures of smokers and nonsmokers are the same.

34. In a 1943 experiment (Whitlock and Bliss, "A Bioassay Technique for Anti-helminthics," *Journal of Parasitology*, **29**, pp. 48–58) 10 albino rats were used to study the effectiveness of carbon tetrachloride as a treatment for worms. Each rat received an injection of worm larvae. After 8 days, the rats were randomly divided into two groups of 5 each; each rat in the first group received a dose of .032 cc of carbon tetrachloride, whereas the dosage for each rat in the second group was .063 cc. Two days later the rats were killed, and the number of adult worms in each rat was determined. The numbers detected in the group receiving the .032 dosage were

    421, 462, 400, 378, 413

    whereas they were

    207, 17, 412, 74, 116

    for those receiving the .063 dosage. Do the data prove that the larger dosage is more effective than the smaller?

35. A professor claims that the average starting salary of industrial engineering graduating seniors is greater than that of civil engineering graduates. To study this claim, samples of 16 industrial engineers and 16 civil engineers, all of whom graduated in 2006, were chosen and sample members were queried about their starting salaries. If the industrial engineers had a sample mean salary of \$72,700 and a sample standard deviation of \$2400, and the civil engineers had a sample mean salary of \$71,400 and a sample standard deviation of \$2200, has the professor's claim been verified? Find the appropriate p-value.

36. In a certain experimental laboratory, a method A for producing gasoline from crude oil is being investigated. Before completing experimentation, a new method B is proposed. All other things being equal, it was decided to abandon A in favor of B only if the average yield of the latter was clearly greater. The yield of both processes is assumed to be normally distributed. However, there has been insufficient time to ascertain their true standard deviations, although there appears to be no reason why they cannot be assumed equal. Cost considerations impose size limits on the size of samples that can be obtained. If a 1 percent significance level is all that is allowed, what would be your recommendation based on the following random samples? The numbers represent percent yield of crude oil.

    A 23.2, 26.6, 24.4, 23.5, 22.6, 25.7, 25.5
    B 25.7, 27.7, 26.2, 27.9, 25.0, 21.4, 26.1

37. A study was instituted to learn how the diets of women changed during the winter and the summer. A random group of 12 women were observed during the month of July and the percentage of each woman's calories that came from fat was determined. Similar observations were made on a different randomly selected group of size 12 during the month of January. The results were as follows:

July	32.2, 27.4, 28.6, 32.4, 40.5, 26.2, 29.4, 25.8, 36.6, 30.3, 28.5, 32.0
January	30.5, 28.4, 40.2, 37.6, 36.5, 38.8, 34.7, 29.5, 29.7, 37.2, 41.5, 37.0

Test the hypothesis that the mean fat percentage intake is the same for both months. Use the **(a)** 5 percent level of significance and **(b)** 1 percent level of significance.

38. To learn about the feeding habits of bats, 22 bats were tagged and tracked by radio. Of these 22 bats, 12 were female and 10 were male. The distances flown (in meters) between feedings were noted for each of the 22 bats, and the following summary statistics were obtained.

Female Bats	Male Bats
$n = 12$	$m = 10$
$\overline{X} = 180$	$\overline{Y} = 136$
$S_x = 92$	$S_y = 86$

Test the hypothesis that the mean distance flown between feedings is the same for the populations of both male and of female bats. Use the 5 percent level of significance.

39. The following data summary was obtained from a comparison of the lead content of human hair removed from adult individuals that had died between 1880 and 1920 with the lead content of present-day adults. The data are in units of micrograms, equal to one-millionth of a gram.

	1880–1920	Today
Sample size:	30	100
Sample mean:	48.5	26.6
Sample standard deviation:	14.5	12.3

a. Do the above data establish, at the 1 percent level of significance, that the mean lead content of human hair is less today than it was in the years between 1880 and 1920? Clearly state what the null and alternative hypotheses are.

b. What is the p-value for the hypothesis test in part (a)?

40. Sample weights (in pounds) of newborn babies born in two adjacent counties in western Pennsylvania yielded the following data.

$$n = 53, \quad m = 44$$
$$\overline{X} = 6.8, \quad \overline{Y} = 7.2$$
$$S^2 = 5.2, \quad S^2 = 4.9$$

Consider a test of the hypothesis that the mean weight of newborns is the same in both counties. What is the resulting p-value?

41. To verify the hypothesis that blood lead levels tend to be higher for children whose parents work in a factory that uses lead in the manufacturing process, researchers examined lead levels in the blood of 33 children whose parents worked in a battery manufacturing factory (Morton, D., Saah, A., Silberg, S., Owens, W., Roberts, M., and Saah, M., "Lead Absorption in Children of Employees in a Lead-Related Industry," *American Journal of Epidemiology*, **115**, 549–555, 1982). Each of these children was then *matched* by another child who was of similar age, lived in a similar neighborhood, had a similar exposure to traffic, but whose parent did not work with lead. The blood levels of the 33 cases (sample 1) as well as those of the 33 controls (sample 2) were then used to test the hypothesis that the average blood levels of these groups are the same. If the resulting sample means and sample standard deviations were

$$\bar{x}_1 = .015, \quad s_1 = .004, \quad \bar{x}_2 = .006, \quad s_2 = .006$$

find the resulting p-value. Assume a common variance.

42. Ten pregnant women were given an injection of pitocin to induce labor. Their systolic blood pressures immediately before and after the injection were:

Patient	Before	After	Patient	Before	After
1	134	140	6	140	138
2	122	130	7	118	124
3	132	135	8	127	126
4	130	126	9	125	132
5	128	134	10	142	144

Do the data indicate that injection of this drug changes blood pressure?

43. A question of medical importance is whether jogging leads to a reduction in one's pulse rate. To test this hypothesis, 8 nonjogging volunteers agreed to begin a 1-month jogging program. After the month their pulse rates were determined and compared with their earlier values. If the data are as follows, can we conclude that jogging has had an effect on the